L08 – Diffusion Models Report

ITAI 2376

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The purpose of this lab was to implement and train a diffusion model capable of generating digits using the MNIST dataset. The main goal was to understand how diffusion models gradually corrupt images with noise, and learn how to reverse the process. I built a U-Net architecture with time and class conditions, and trained it to visualize forward and reverse diffusion processes using MNIST. I also used CLIP to evaluate the generated digits and see how they matched the descriptions. In this lab, I learned about diffusion, noise, sampling, and quality evaluation.

1. Understanding Diffusion

What happens during the forward diffusion process is the gradual corruption of an image by adding small amounts of noise. At the beginning, the image is clearer, but as it progresses, the structure fades, edges blur, and the fine details disappear. The reason why we add noise gradually is because if we added all the noise in one step, our model wouldn't reconstruct the image. When we gradually add noise, our model can break the problem into small steps, can learn local corrections, training is stabilized, and we can ensure that the reverse process is smooth. During my lab, most digits became recognizable around 40-60% of the denoising process. I was able to recognize digits like 7 and 0, but digits like 5 and 9 were more difficult, so it does vary by digit.

2. Model Architecture

The U-Net architecture is well-suited for diffusion models because the downsampling path captures global context, the upsampling path reconstructs the fine details, and skip connections preserve high-resolution information. Skip connections are important because they directly pass features from the encoder to the decoder in the model. In the model, each DownBlock outputs a skip tensor and the UpBlock receives it, and the decoder combines coarse global features with the final local details. The way class conditioning works is by converting the digit label to a one-hot vector, the EmbedBlock transformed into a 128-dimensional embedding, and reshape it, and then added to the bottleneck features of the U-Net, and finally, the mask allows conditional or unconditional generation.

3. Training Analysis

The loss value tells us that the model has good denoising ability, is more accurate when reconstructing, and has higher quality. The images in the first epoch looked blurry and didn't have much structure. During mid-training, I started to recognize some shapes, and at the end of training, the edges and curves became clearer. I was able to observe a gradual improvement with each epoch during training. The reason we need time embedding in diffusion models is because

it tells us how much noise remains at each step. In early timesteps, the heavy noise is removed, in middle timesteps, the structure is restored, and in late timesteps, the details are refined. Without embeddings, the model would treat all the steps the same, making our model fail to denoise properly.

4. CLIP Evaluation

The CLIP scores tell us if the digits match "a handwritten number X," "a clear, well-written digit X," and "a blurry or unclear number." The images that showed a high score in my lab were 1, 7, and 0, and numbers like 5 and 8 showed lower scores. The images that looked blurry were flagged with high blurry scores. I think the reason why some digits are easier for the model to generate is because their shape is simpler, they have stronger global structure, and low variance. The shapes of more complex digits require fine details that get lost in noise. The way CLIP scores can be used to improve the generation process is by using CLIP-guided diffusion to push the generated digit towards the target concept, improving the consistency.

5. Practical Applications

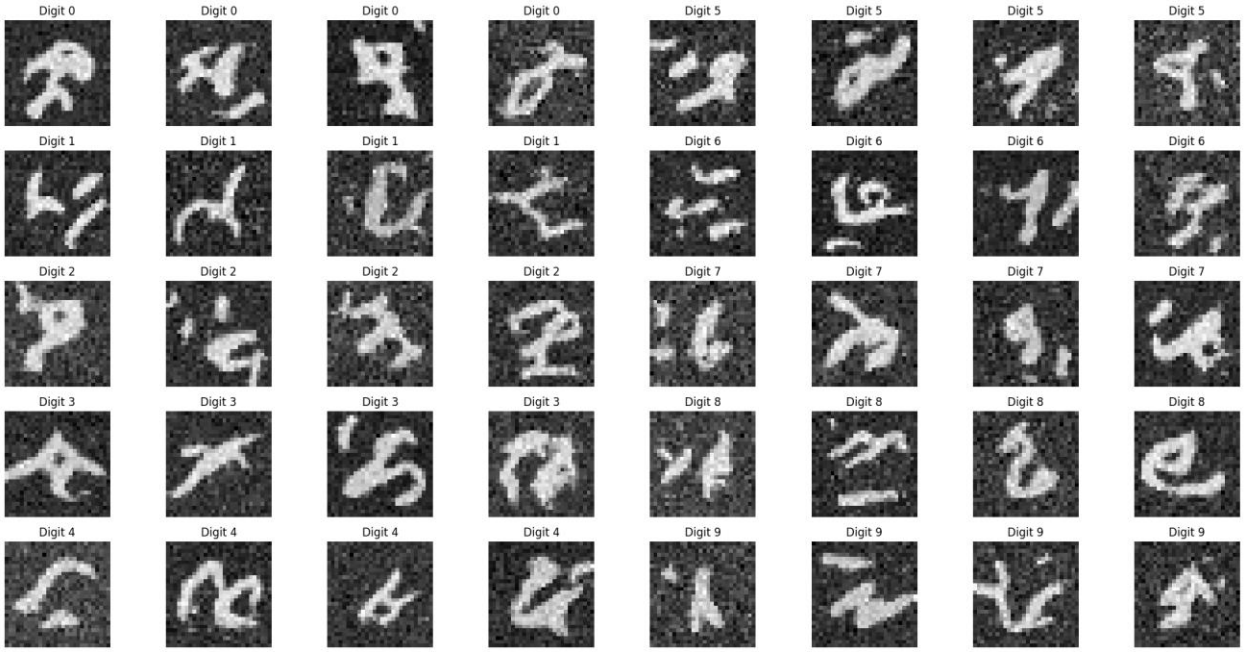
This type of model can be useful for things like image restoration, medical imaging reconstruction, handwriting synthesis, etc. The limitation of our model includes low resolution, slow samplings, limited diversity, and trained on a simple dataset. If I continue working in this project, I would add classifier-free guidance for more consistent digits, add Exponential Moving Average for more stable samples, and train the model on a higher-resolution dataset to improve details.

Additional Observations

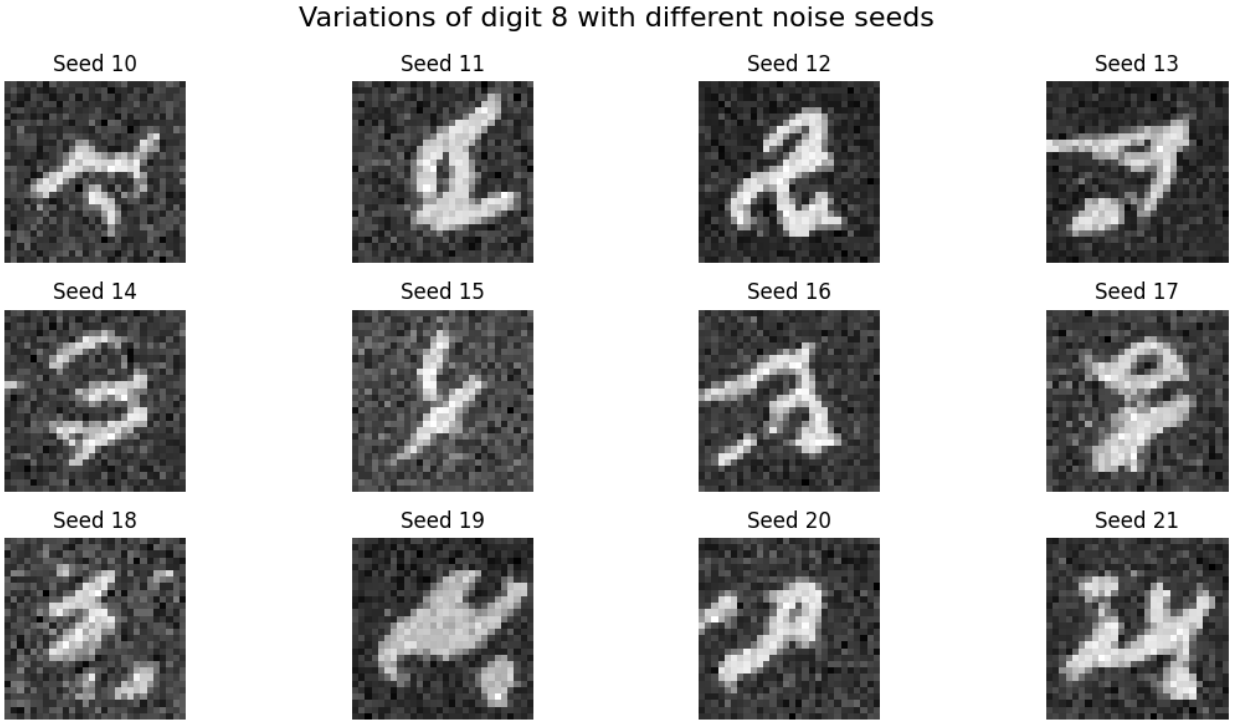
When I was working in this lab, some of the things that I notice were that the digit complexity affects quality; simple digits are easier for the model to generate. The denoising process is intuitive. When we reach like half the reverse diffusion steps, I was able to see some shapes. When I used CLIP evaluation, I noticed that my judgment and CLIP scores matched for the most part. I notice how different noise seeds produce variations of the same digit; some have different curvatures and thicker strokes. During training, I observed the improvement though each epoch, the first "digits" looked like noisy blobs, but at the end the digits had sharper edges and consistent shapes. I also realized a big mistake on my part, when I was literally done with the lab, I notice that the images my model was generating didn't look right at all, so I check the whole lab and find out that I was using the wrong dataset. Instead of MNIST, I was using Fashion-MNIST, and that was the reason my digits didn't look like digits, but I was able to correct my error and get better results.

Images generated during step 6

Generating 4 versions of number 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

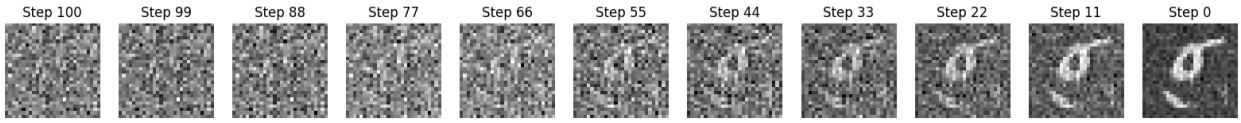


STUDENT ACTIVITY: Generating numbers with different noise seeds

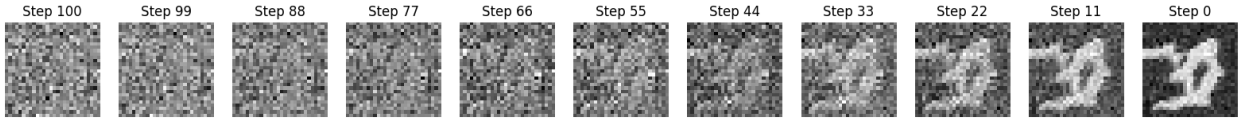


Images generated during step 7

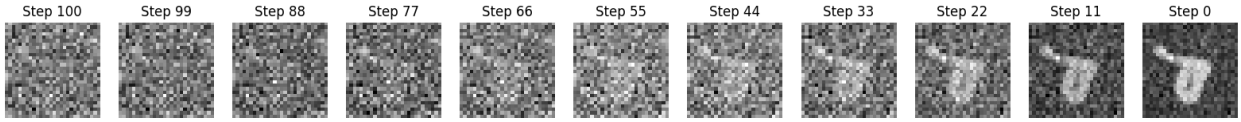
Generating number 0:



Generating number 3:



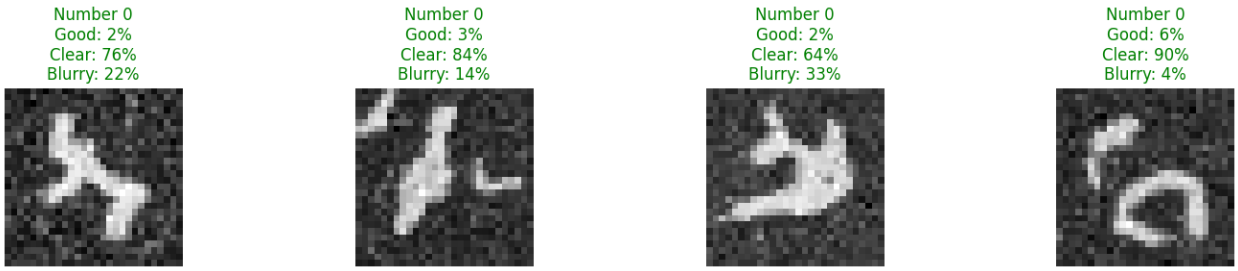
Generating number 7:



Images generated during step 8

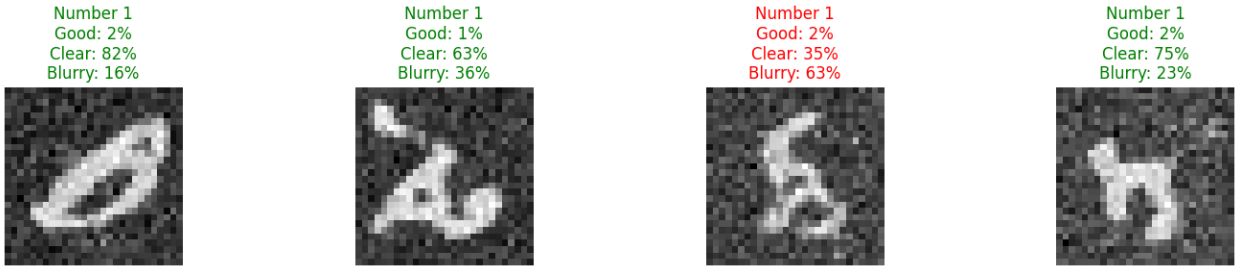
Generating and evaluating number 0...

Generating 4 versions of number 0...



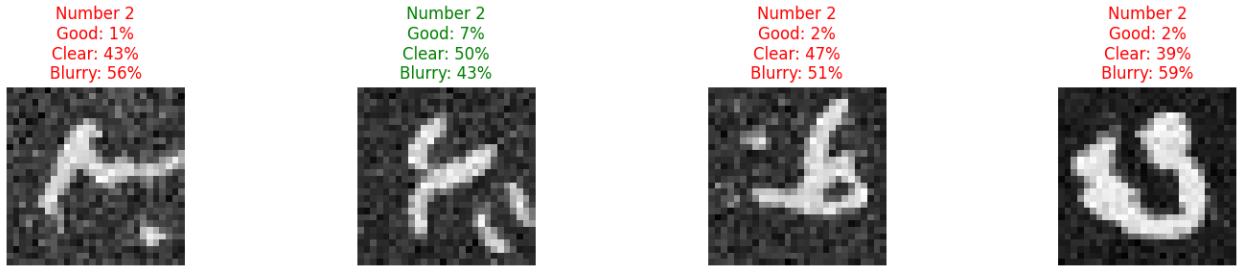
Generating and evaluating number 1...

Generating 4 versions of number 1...



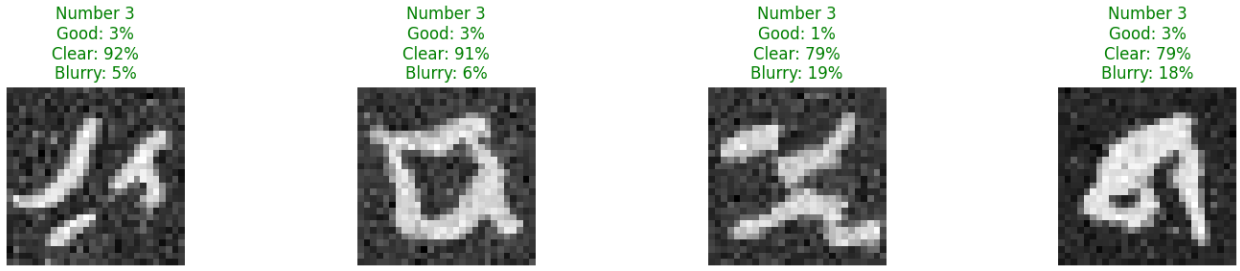
Generating and evaluating number 2...

Generating 4 versions of number 2...



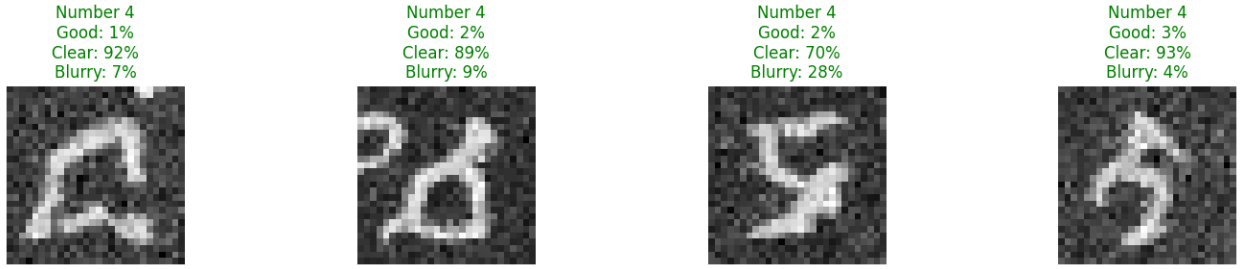
Generating and evaluating number 3...

Generating 4 versions of number 3...



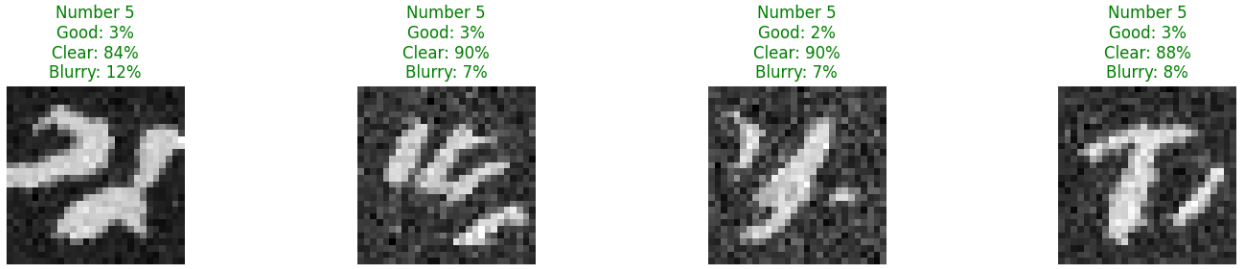
Generating and evaluating number 4...

Generating 4 versions of number 4...



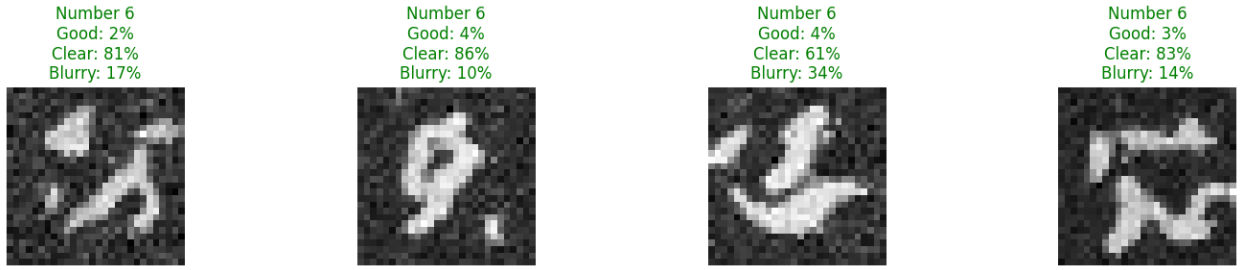
Generating and evaluating number 5...

Generating 4 versions of number 5...



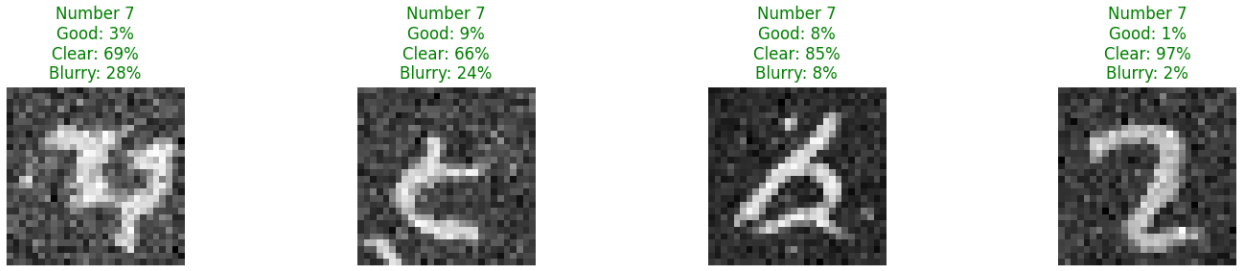
Generating and evaluating number 6...

Generating 4 versions of number 6...



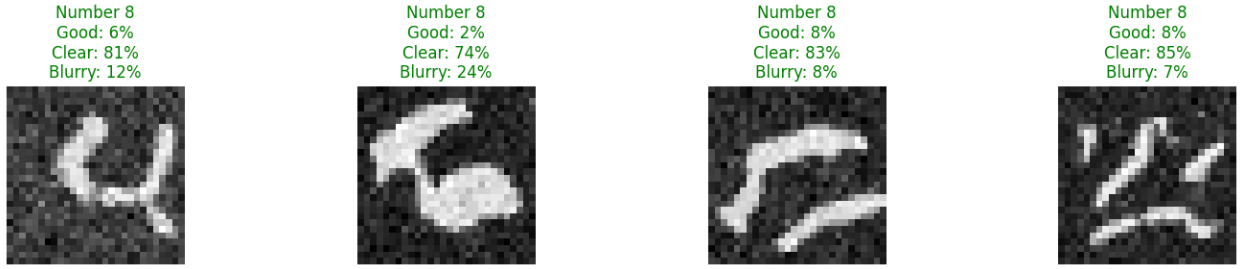
Generating and evaluating number 7...

Generating 4 versions of number 7...



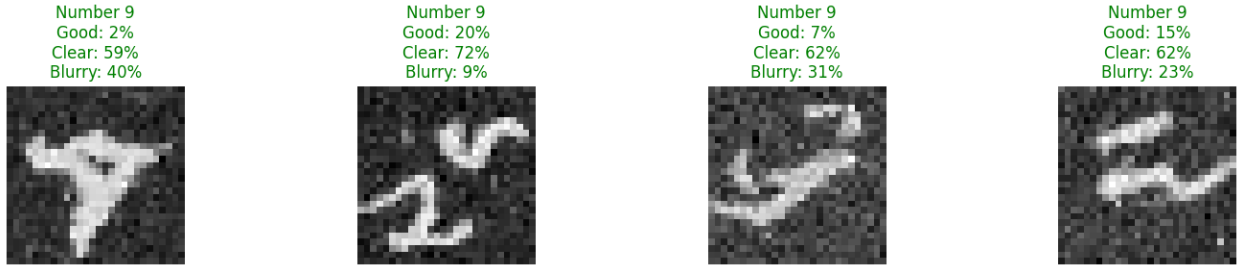
Generating and evaluating number 8...

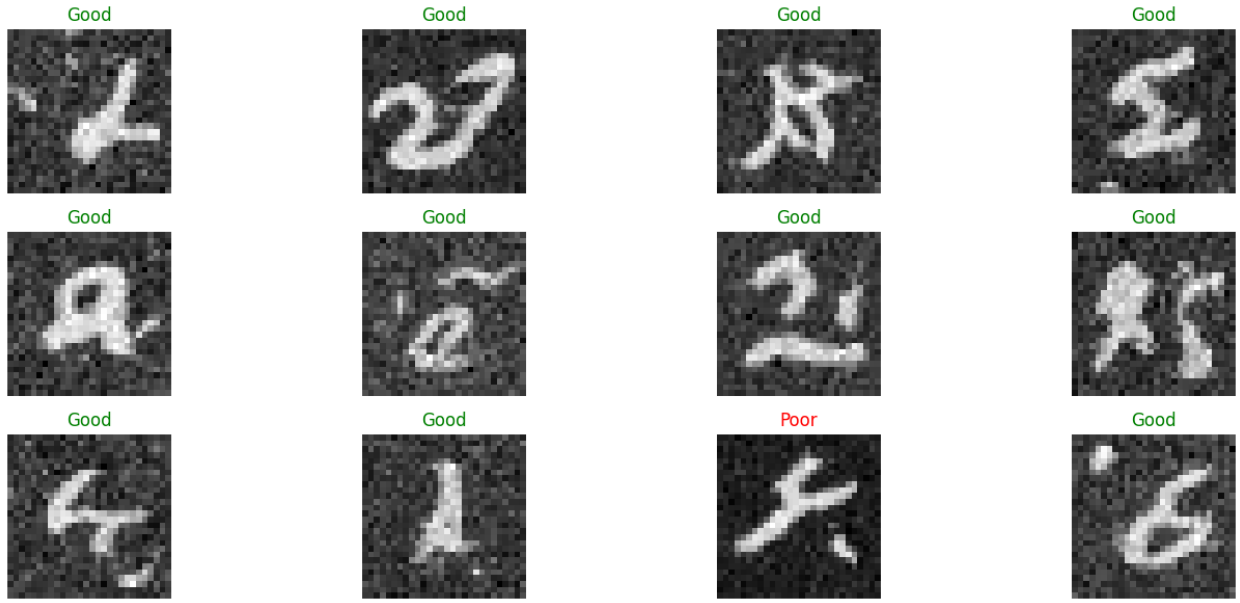
Generating 4 versions of number 8...



Generating and evaluating number 9...

Generating 4 versions of number 9...



Generating 12 versions of number 6...

CLIP judged 91.7% of the generated '6' digits as good quality.